

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M. Tech)

Semester – I University Examination January 2010

M. Tech. (CAD/CAM) (Theory)

ME – 701 Computer Aided Design

Date: 03. 01. 2011, Monday

Time: 10:30 p.m. to 01:30 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.
3. Use of PSG Design data book is permissible.

SECTION- I

- Q1 (a) Why can't a vector scan display produce as many colours as a raster scan display? 2
- (b) How does the CAD help to reduce product development cycle? Explain. 3
- (c) State the modelling facilities that are expected in good solid modelling software. 3
- (d) By giving suitable examples explain the various techniques used for solid modeling. 5
- Q2 (a) What is Window? What is View Port? 3
- (b) What is scan conversion? Explain scan conversion of a line. 3
- (c) A line joining point P (12, 12) and Q (20, 18) is to be drawn using Bresenham's algorithm. Find the coordinates of pixels starting from point P. 5

OR

- Q2 (a) What is clipping? Explain Point and Line Clipping. 3
- (b) Explain: "Bresenham's algorithm is more efficient than the DDA line generation algorithm". 3
- (c) Write Bresenham's circle drawing algorithm with (x_c, y_c) as center and radius r . 5
- Q3 (a) What is homogenous coordinate system? Why is it preferred in computer graphics? 3
- (b) Compare geometric transformation with coordinate transformation. Also explain composite transformation. 3
- (c) A triangle PQR has its vertices at P (4, 2), Q (8, 2) & R (6, 5). It is to be rotated in CCW direction about the point (6, 5) through 90° . Find the new position of the triangle. 5

OR

- Q3 (a) Write the matrices for the following 2-D transformations: 3
- (i) Scaling (ii) Rotation (iii) Shearing.
- (b) Obtain 3D rotations for cases: 3
- (i) Rotation about x-axis followed by equal rotation about y-axis.
- (ii) Rotation about x-axis followed by equal rotation about y-axis.
- Hence prove that 3D rotations are not commutative.
- (c) Show that the transformation matrix for reflection about the line $Y = -X$ is equivalent to reflection relative to Y axis followed by counterclockwise rotation by 90° . 5

SECTION- II

- Q4 (a) Write down the steps for generation of solid model of a Hexagonal Bolt using any solid model software. 4
- (b) Explain parametric continuity conditions for curves. 4
- (c) Explain feature based models. How are those constructed? 4

- Q5 (a) Explain rotary mode and plane mode of scanning techniques & list out different reverse engineering software's. 5
- (b) The coordinates of four control points relative to a current WCS are given by $P_0 = [2 \ 2 \ 0]^T$, $P_1 = [2 \ 3 \ 0]^T$, $P_2 = [3 \ 3 \ 0]^T$ & $P_3 = [3 \ 2 \ 0]^T$. Find the equation of the resulting Bezier curve. Also find points on curve for $u = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \text{ \& } 1$. 6

OR

- Q5 (a) What is reverse engineering? Justify its use. 3
- (b) Explain various properties of Bezier curves. 4
- (c) Calculate the parametric midpoint of Hermite cubic curve that fits the points $P_0 = (1, 3)$, $P_1 = (7, 2)$ and the tangent vectors $P'_0 = (10, 8)$, $P'_1 = (6, 0)$. 4
- Q6 (a) Write a short notes on the following: 6
- (i) GKS (ii) NURBS
- (b) Write an interactive program and algorithm for four bar chain mechanism. 6

OR

- Q6 (a) Discuss the need for standardization in computer graphics & list the various CAD standards. 6
- (b) Prepare an exhaustive algorithm for design of connecting rod of an IC engine 6